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TITLE: *A Network Model of R&D, Competition and Intellectual Property*

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Abstract

R&D is vital for the continued advancement of humanity's welfare. However, the exact mechanisms that incentivize it under competitive conditions are still subject to significant academic debate. This work develops a network model focused on product development within varied competition scenarios under simple premises. Firms can engage in original research or try to imitate and reverse-engineer their competitor's products, with which they will compete for market share. The results show significant realism, being, for the most part, consistent with similar models found in the economic literature, and shedding light on important points of controversy, such as the effects of competition levels and Intellectual Property protections on firm's R&D efforts. The present analysis supports the notion that increased competition may reduce R&D efforts by reducing prospective profits for innovating firms, and a relaxation of patent enforcement can enhance research efficiency by mitigating duplicative costs, while also reducing the distance in the level of development between competing firms. Despite the clear results, the work acknowledges several critical shortcomings of the model and proposes further refinements that may improve the validity of its findings.

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1. Introduction

The development of new and better products is a key factor for the continued improvement of consumer welfare. Not only is there a human desire for more and better products, but also trends and tastes change over time, motivating firms to adapt and offer better matches to buyer's desires. Product developments can come in many forms: bigger, smaller, stronger, lighter, disposable, longer-lasting, or simply different. They can also benefit other firms, in the form of better, more useful, and more precise tools and machines, enabling the creation of new inventions, never seen before. But significant breakthroughs are few and far in between. Instead, the bulk of the advancement is done through incremental steps: building over what already exists, little by little. In order for this incremental process to take place, two prerequisites must concur. The first is access to a knowledge base, meaning, something to build upon. And the second is the incentive, the actual motivation that brings the effort forward.

Let's address the first prerequisite: knowledge. It can take many forms, technical or practical, theoretical, and more. But what they have in common is that they spread through interaction between agents. The latest development in modelling this diffusion across a network of agents is network modelling, which mathematically simulates an interconnected grid of agents, where the links between them influence the spread of knowledge, among other effects. This framework allows the recreation of complex topologies of interactions beyond the classic microeconomic models.

The second prerequisite, the incentive, is also vital, because, without an agent performing the action, no progress can take place. Mere curiosity helps here, where an intrinsically motivated agent carries out the task for its own good. But this is clearly insufficient to satisfy the demands of millions in a world increasingly driven by technology. In his classic work, "The Wealth of Nations", Adam Smith puts it succinctly:

"It is not from the benevolence of the butcher, the brewer, or the baker, that we expect our dinner, but from their regard to their own interest. We address ourselves, not to their humanity but to their self-love, and never talk to them of our own necessities but of their advantages." – Adam Smith, 1776.

It is this, the strong motivation provided by the pursuit of profit, that brings firms and competition into the matter. By operating for its own interests in a market-driven economy, firms can provide the necessary workforce to carry out R&D.

Markets and competition, however, are not simple matters. The alignment in incentives between firms and consumers is often complex, more so when considering differing levels of quality, and even more regarding something for what there is not yet a market: a new invention.

It is with these matters that network models can help us: not only on the knowledge side, but also on the competition side. Since both factors can be modelled using networks, they present themselves as the ideal tool for an informative economic simulation.

However, there is yet another level of complexity on top. Since the 19th century, and with increasing frequency, firms have been relying on patents and intellectual property protection to keep the proceeds of their efforts. Where secrecy and exclusive skill had been the foundation of the innovative industry of the past, a whole bureaucracy is now in effect, tasked with administering and enforcing the rights over discoveries and inventions.

This work aims to explore the differential effects of the presence and absence of intellectual property rights in the R&D efforts and profits of firms immersed in varied competitive scenarios. Based on the works in network models by Bramoullé & Kranton (2006), Bramoullé et al. (2014) and Goyal & Joshi (2006), I developed an R&D competition network model where firms compete to develop the highest-quality products in order to capture market share. With a focus on simplicity and tractability, multiple scenarios are tested, with diverse connections between firms, and the strategic decisions of firms regarding research and imitation efforts are observed.

The outputs of the model will then be analyzed and critiqued for realism in order to obtain insights and add evidence to several controversial debates within the academic literature in economics. In particular, the questions that this work aims to answer are the following.

1.1 Research questions

1. Under what conditions do firms exert effort in original development?

Since a faster technological advancement has the potential to increase humanity's quality of life at a greater pace, it would be interesting to find ways to foster it.

2. How does patent law infringement affect the quality of products in competitive markets?

Better quality is a fundamental aspect of both industrial and consumption goods, which is highly regarded by buyers. In this model, quality is an indication of the level of development a firm has achieved for a certain product.

3. Would the elimination of patent protections increase efficiency?

Many prominent voices have raised concerns about the effectiveness of the current legal regime in promoting technological progress. While patents are supposed to incentivize firms to invest in R&D, critics point out that it can also delay it by preventing other firms from using the knowledge as input to investigate new findings.

1.2 Overview of Content

First, the relevant academic literature is reviewed and summarized. This will be useful later to assess the credibility of the model's results. Then, the model itself is formally defined and described, with explanations for all its variables, parameters and equations. An overview of its two Maple implementations follows, which presents insightful graphs and descriptions of its workings and characteristics. Both implementations can be found in the Appendices B and C, respectively. A description of how to set up the model is given, in case the reader wishes to replicate it.

Subsequently, the scenarios incorporated in the model are described, followed by a verbal description of each of their results, which can be found in Appendix A. The results from individual scenarios are then compounded and synthesized into a list of useful insights, detailing the general tendencies in the model's behavior. Each of the insights is then compared with the relevant literature in order to critique its realism and, consequently, the validity of the findings. With this information, the questions outlined above are answered according to the model's results, followed by a critical acknowledgment of its limitations. Finally, the work ends with some concluding remarks.

2. Theoretical Framework

2.1 Competition and Innovation

There is no consensus on whether the level of competition in an industry is beneficial or detrimental to the aggregate innovative effort of the firms involved. (Beneito et al., 2014). The debate originated with the theory of Schumpeter in *Capitalism, Socialism and Democracy* (1943), which stated that a temporary monopoly provides the crucial incentives for firms to undertake the risks associated with innovation, contradicting the prevailing view at the time. Years later, Arrow (1962) argued back that a pure monopoly has less incentives to invest in R&D than a competitive firm. This gave rise to Arrow's replacement effect, which states that higher initial profits for the incumbent reduce the value of the innovation. (Boone, 2000).

On the empirical side, some sources support the idea that high levels of competition foster innovation and investment in R&D, such as Aghion et al. (2005), and that they increase the amount of patents filed (Boldrin & Levine, 2013). Meanwhile, other studies, such as Dasgupta & Stiglitz (1979), indicate a positive association between industry concentration and R&D investment. Overall, many studies acknowledge the complexity of the relationship, such as Gilbert (2006). In particular, a common finding is a non-monotonic relationship in the form of an inverted-U, where there is a "right amount" of industry concentration that maximizes innovation. (Aghion et al., 2005; Beneito et al., 2014; Levin et al., 1987).

2.2 Types of Innovations

R&D developments can be separated into 2 types of innovations: process and product. (Boone, 2000). Process innovation is defined as any improvement in the production method. In economics, it is typically modelled as a reduction in the cost of production of the product. (Gilbert, 2006). On the other hand, product innovation is defined as a change in the form of the final good which yields either an improvement in its quality (or perceived utility by the consumer) or that differentiates the product from its standing competition. (Glass & Wu, 2007).

2.3 Competition and Quality

The most common finding in the academic literature is that competition either harms quality or they follow a more complex relationship. (Gilbert, 2006). Leeland (1977) provides some insight into the matter by suggesting that monopolists tend

to underprovide quality, but cautioning that it does not necessarily follow that competition will lead to greater quality. In a model by Shaked and Sutton (1982), firms in a duopoly differentiate their products by quality as a strategic move to relax price competition. However, if more firms are present, the differentiation tends to disappear, leading to a “flat” level of quality for all firms.

Firms make decisions about the price, quality and quantity of their products. Kranton (2003) provides a theoretical model where price competition can eliminate the profits necessary for firms to maintain a reputation for high quality goods. This effect is especially likely when quality is not observable by consumers. Therefore, in a competitive environment, a perfect equilibrium where firms produce high-quality goods may not exist. However, Aghion et al. (2005) do find that competition tends to exert pressure on the less productive firms, either precipitating their exit, or forcing them to increase their quality to a minimum standard in order to compete.

2.4 Intellectual Property Protection

The protection of intellectual property (IP) can be traced back as far as 1474, in the Republic of Venice. Entrepreneurs were granted exclusive rights for 10 years over their inventions and the technologies they imported. The practice was imitated by other European countries in the hopes of promoting economic development. (Moser, 2016).

The intent behind patents is to resolve the tension that exists between the cost of innovation and the widespread diffusion of its benefits. It offers the innovator a temporary monopoly over its invention in exchange for its public disclosure, spreading its benefits when the patent expires. (Levin et al., 1987). Along the same lines, Gilbert (2006) stated that studies showing a higher social than private return to research suggest that policies that promote innovation can be beneficial to society at large.

Several papers discuss the merits of IP protection, often stressing its negative effects when applying them too stringently. In this regard, Glass & Wu (2003) pose that weak protection of IP rights actually increases the aggregate rate of innovation. Moser (2016) explains that extending the length of patents can encourage innovation by granting inventors stronger property rights in their ideas, but they can also discourage innovation if they reduce the payoffs of later innovators who rely on previous inventions as inputs for their work. She also states, based on historical evidence from exhibitions of inventions, that the

absence of patent protection did not prevent innovation in countries without patent laws.

Arrow argued in 1962 that, with exclusive intellectual property rights, a firm in a competitive industry had a greater incentive to invest in R&D than a monopolist, while observations by Boldrin & Levine (2013) are consistent with theories of innovation that emphasize first-move advantage as the main driver of innovation, favoring competition and directly contradicting the Schumpeterian theory of monopolistic innovation. However, because markets vary widely in their level of competition, some papers have developed insights on how IP protection interacts with competition. Gould & Gruben (1994) argue that patent enforcement is more likely to foster economic growth in open economies, since, in closed regimes, the competitive framework is inadequate to stimulate innovation. They also point out that firms in closed-trade regimes typically find it more profitable to copy foreign technology than to innovate.

2.5 Secrecy & Reverse Engineering

Before the advent of patents, innovators typically relied on secrecy in order to appropriate the profits of their inventions. In fact, when patent protection is weak, inventors may choose to keep their new technologies secret, so that competitors may not copy their discoveries. (Gilbert, 2006). However, with the advancement of science, new technical tools became available that allowed rivals to reverse-engineer inventions, diminishing the effectiveness of secrecy. This phenomenon was of particular importance in the chemical industry of the mid-nineteenth century, shifting the industry towards an intensive use of patents to protect their inventions. (Moser, 2016).

Patent protection offers a significant advantage over secrecy: it allows for the widespread diffusion of the newly acquired knowledge, which would otherwise be kept confidential. (Boldrin & Levine, 2013). However, where secrecy is itself a method of enforcement, on the flip side, patents require the use of force to stay in effect. This presents a key vulnerability in environments with uncooperative authorities. Foreign governments, especially, may have incentives to breach the IP protection of foreign investors: Gould & Gruben (1994) find that closed-trade regimes typically find copying foreign technology more profitable than innovating. In particular, Moser (2016) points out that the US patent system may have encouraged economic development by reducing the costs of copying foreign inventions.

3. Model

3.1 Formal Definition

The set of **players** is the set of N firms denoted by $I = \{1, 2, \dots, N\}$.

For each firm $i \in I$, a **strategy** σ_i consists of the triplet $(x_{d,i}, x_{e,i}, T_i)$, where:

- $x_{d,i}$: Firm i 's effort in product improvement through original development, where $0 \leq x_{e,i} \leq Q_{T,i} - Q_{0,i}$
- $x_{e,i}$: Firm i 's effort in product improvement through reverse-engineering, where $x_{e,i} \geq 0$.
- T_i : Firm i 's chosen target firm for reverse-engineering, where $T_i \in I$.

A **strategy profile** is a combination of strategies for all firms: $\Sigma = (\sigma_1, \sigma_2, \dots, \sigma_N)$.

The **payoff** for each firm i is its profit, denoted by $U_i(\sigma) = P_i(\sigma_i, \sigma_j)$, where σ_j denotes the strategies of all firms other than i .

Players choose their strategies **simultaneously** with **complete information**.

3.2 Description

In this model of R&D competition, the players are firms that compete with each other for sales in their respective markets, based on product quality. This product quality can be increased by reverse-engineering their rival's technology and through original research and development, but it cannot modify the competitive relationships between firms.

Each company has a determined starting quality ($Q_0 \geq 0$) and symmetric and immutable links to other companies ($0 \leq L_{ij} = L_{ji} \leq 1$), signifying the strength of the competition between them. This link influences both how sales are distributed and how much reverse engineering costs.

Reverse engineering and research efforts lead to product improvement, increasing both the company's market share, as well as the market's size. Firms compete for market share, which is dictated by their product's quality. Cost of production is not modelled, and neither are price and quantity, which are assumed to increase together with the quality of the product. All these factors are condensed into net sales.

Each node in the network represents a firm, with the lines or edges being the closeness or level of competition between firms. This closeness is defined in an abstract manner, encompassing all possible factors that could make competition more direct between firms, including any interplay with geography, distance, product substitutability, regulatory, language or accessibility considerations, as well as connecting infrastructure and marketing efforts.

The model simulates any kind of product which goes through a development and launch phase. Companies can only copy from already launched competitor's products, while they make efforts to increase their quality to compete at the time of the next launch date.

3.3 Exogenous variables

Q_0 : Product quality at the beginning of the current stage.

D : Parameter of research & development costs.

E : Parameter of reverse-engineering costs.

L : Link strength between companies. $L \in [0,1]$

M : Market size scaling parameter.

N : Number of companies in total.

3.4 Endogenous variables

Q : Product quality at launch.

Q_T : Product quality of the target firm for reverse engineering, at the beginning of the current stage. Target T chosen endogenously given list of exogenous Q_0 .

x_d : Effort in product improvement through original development.

x_e : Effort in product improvement through reverse-engineering or copying.

S : Income from net sales in the launching phase.

P : Company profit net of costs.

3.5 Equations

- **Product quality** is given by the firm's starting quality Q_0 added to the reverse-engineering (x_e) and development (x_d) efforts.

$$Q_f = Q_0 + x_d + x_e$$

$$Q_0 \geq 0$$

Quality is taken as an additive, unidimensional measure, where efforts directly and linearly contribute to it. It cannot be negative, but it can be 0, meaning absolute absence of a product, and up to infinity.

- The **cost of development** is a second-degree function of the development effort that takes into account the reverse-engineering effort previously performed. This cost can be graduated linearly using the parameter D .

$$C_d = \int_{x_e}^{x_e + x_d} D \cdot (1 + q) dq = D \cdot X_d \cdot \left(1 + X_e + \frac{X_d}{2}\right)$$

$$X_d \geq 0$$

This formula highlights the increasing marginal cost of achieving significant advancements in a limited amount of time under the technological constraints of the starting point Q_0 . The linear component is meant to compensate for the very small values that otherwise result when $0 < x_d < 1$.

- The **cost of reverse-engineering** is given by the amount of quality copied from a target firm T , which can be scaled using the parameter E . The maximum amount a firm can copy is given by the difference between their starting quality and the target firm's starting quality, and the cost increases linearly for firms with a lower link strength. Unlinked companies ($L=0$) cannot copy from each other, and only companies with higher quality at start can be copied from.

$$C_e = \int_{Q_0}^{Q_0 + x_e} \frac{E}{L_{i,T}} dq = \frac{E}{L_{i,T}} (Q_0 + x_e - Q_0) = \frac{E}{L_{i,T}} \cdot x_e$$

$$0 \leq L \leq 1$$

Although reverse-engineering can be a difficult and non-deterministic process, here it is modelled as additive for simplicity. Its linear nature represents how it

is much easier to copy than to accomplish original development, especially for great advancements.

Meanwhile, the link factor represents the added difficulty of obtaining information from a product that may be very different in nature from the firm's own, that comes from a distant country with a different culture and language, or that may be very difficult to acquire in the firm's foreign market. Products from firms not in competition with firm i cannot be reverse engineered, for they either bear no resemblance whatsoever with firm's i product, or are impossible to obtain in firm i 's market.

- The **net sales** obtained by each firm are given by the following formula. A global parameter M scales the resulting amount.

$$S_i = M \cdot \frac{Q_i^2}{Q_i + \sum_{j \neq i}^N L_{ij} \cdot Q_j}$$

This is the main piece of the model, and it incorporates 2 distinct mechanisms:

First, market size grows linearly with the final Q of the firm at hand. This represents market growth due to more customers being attracted to a better product regardless of competition.

Second, a distribution mechanism awards sales to each company given their proportion of the total quality in the market. Due to the qualitative nature of the model, there is no set standard that would guide customers to opt for one product over another based on quality alone. Many distributions could work here, but more complicated formulas tended to make the model notably unstable, and so robustness was preferred over realism. Therefore, a simple linear distribution was chosen.

Note that this formula is firm-centric, that is, it applies to each firm independently, not to the market as a whole. Therefore, each firm only feels competitive pressure from other firms to which, and in the proportional amount by which, it is connected to.

- The **profit** each firm receives at the end of the simulation is effectively the payoff of the game. It is calculated as its net sales minus its reverse-engineering and development costs.

$$P = S - C_d - C_e$$

This is a simple accounting equation that highlights the balance between costs and revenues for an optimal payout.

3.6 Implementation

Two Maple worksheets were developed to run the model, which can be found in the appendix.

First, a simple 2-firm version was created and the equilibrium found through algebraic manipulation. This version contains code that allows for the creation of 3D graphs showing relevant aspects of the interaction such as payoff functions and best responses to the competitor's actions.

A toy model without reverse-engineering was set up in this early version, with parameters:

- $Q_{0,1}=10$
- $M=100$
- $x_{e,1}=0$
- $Q_{0,2}=1$
- $D=1$
- $x_{e,2}=0$
- $L_{12}=0.75$
- $E=N/A$

Which presents the following payoffs as a function of $xd1$ and $xd2$.

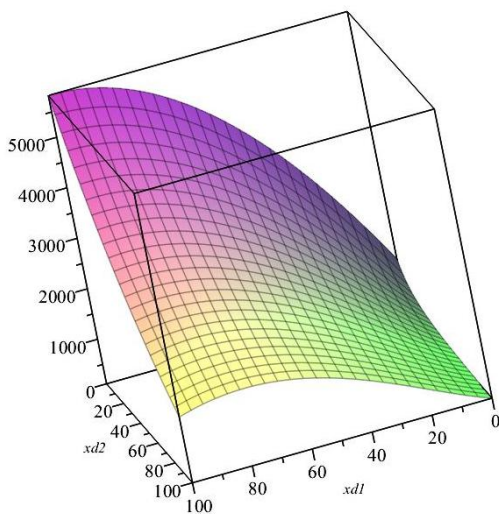


Figure: Payoff function of firm 1 in the toy model

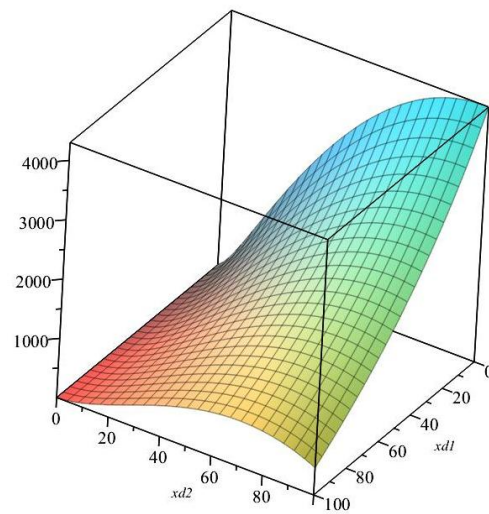


Figure: Payoff function of firm 2 in the toy model.

Then, a more advanced worksheet was developed, which supports N-firms and incorporates an iterative numerical algorithm in order to find the Nash equilibrium. Due to the complexity of the interaction and the numerical nature of the solver, it does not support 3D graphs like the simpler version does, but it does incorporate a network graph builder which shows the topography of the network, with nodes

for firms and connecting links with labels indicating the competition strength between each.

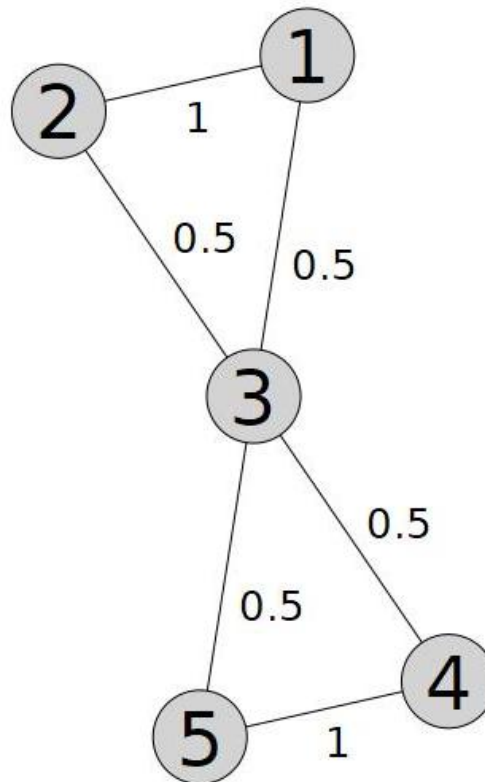


Figure: Example of network graph.
Scenario 3 with 5 firms.

This more refined model was then used to test multiple scenarios, which are described below, and their outputs can be found in appendix A.

3.7 Setup

The model is initialized by inputting the number of firms N , the starting quality of each firm ($Q0x$), and the parameters D (Dc), E (Ec) and M (Ms). Then each individual link is inputted as Lxy , where self-links (Lxx) are not necessary, and any links not provided are assumed to be 0. Extra unused links are ignored.

The program then proceeds to run, generating a network graph of the system as described by the parameters, and processes them according to the pre-inputted formulas.

Lower in the worksheet, the user can tune the iteration parameters, which are: max iterations (*maxIt*), tolerance between iterations (*tol*) and the damping factor (*df*). However, modifying these values is typically not necessary unless the defining formulas are changed, or some very complex or obscure system is simulated, which may cause issues in the iteration process.

Further below, a varying number of warnings of the type “*no iterations performed as initial point satisfies first-order conditions*” can appear, but these are normal. Right after those, an output shall appear where it indicates whether the system has successfully converged, and how many iterations it took.

At the bottom of the worksheet, the system will output the results of the simulation in the form of a table. The presentation of the table, as well as the output variables, can be modified according to the user's wishes, but it is not necessary.

3.8 Parameters

For all presented scenarios, the parameters of the simulation are set as follows:

- $D = 1$
- $E = 1$
- $M = 100$
- $Tol = 10^{-4}$

3.9 Scenarios

- Single Market: Firms find themselves in perfect competition with each other. (Appendix A1).
- Two Markets: Two clusters of firms, which are in perfect competition within the same group, but only have weak links with firms from the other cluster. (Appendix A2).
- Closed Silk Road: A scenario inspired by the Silk Road to the Orient. It consists of a series of markets in succession, where each market is connected only to its neighbors. (Appendix A3).
- Open Silk Road: As in the previous scenario, markets are aligned in succession. Although all companies possess links to each other, link strengths fall significantly the farther away their markets are from each other. (Appendix A4).
- Global Market: A full network of interconnected firms, which are only in imperfect competition with each other. (Appendix A5).

4. Results and Analysis

4.1 Results: Single Market

See appendix A1.

- $N=1$

Starting with a low quality and no competition, an initial monopolist still chooses to exert a significant amount of development effort. (Table A1.1).

- $N=2$

Adding a direct competitor reduces both profits and margins for the market leader, while quality is slightly lowered. This new competitor exerts a moderate amount of development effort, but falls significantly short of the leader, obtaining a very small profit with a thin margin. (Table A1.3).

With reverse-engineering, the position of the new contender is improved. Now it does not engage in development effort, but instead maximizes its reverse-engineering effort, reaching the same quality that the incumbent leader had at the start. Although it obtains a very wide margin due to the low costs of copying, its profits are notably low compared to the market leader. The leader itself is, however, hurt by the illegitimate competition, obtaining somewhat lower profits and margin, and reaching an even lower final quality, although only slightly. The joint profit is reduced by the addition of a competitor, but this effect is moderated slightly when allowing for reverse engineering. (Table A1.4).

- $N=3$

With 3 firms, we observe a similar dynamic between the first two firms. However, this time the third contender does not manage to find space in the market for itself, foregoing any product development, and ultimately failing to obtain any significant profit. (Table A1.6).

Allowing for reverse engineering permits the third contender to obtain an easy but small profit with a wide margin, without engaging in any original development. We observe a similar reduction in the leader's quality, profit and margins as before. Similarly, the reduction in joint profit is also moderated by the addition of reverse engineering. All firms target the leader and exert the maximum amount of copying effort. (Table A1.7).

- N=4

In the simulation with 4 firms, we have added a small amount of starting quality to the third firm, as opposed to the newest contender, who starts from scratch. This small change propels it to exert some reverse-engineering effort, although the resulting profits are low with tight margins, well below the second firm. As before, the last firm fails to develop any additional quality and obtains negligible profits. (Table A1.9).

With reverse engineering, the last 2 firms show very similar results with no development and comparable profits and margins. Both the leader's as well the second firm's final quality and profits are hurt by the competition, although the second firm's development effort falls disproportionately, leading to a higher margin than before. (Table A1.10).

4.2 Results: Two Markets

See appendix A2.

- N=2

With only 2 firms, each has a clear presence in its own market, both engaging in significant amounts of development. The second firm, however, is unable to keep pace with the advanced start of the leader and obtains significantly lower profits with much tighter margins. (Table A2.2).

In this case, little changes with the allowance of reverse engineering. The leader's final quality, profit and margin are unchanged, as is the final quality of the second contender. The latter, however, substitutes its development effort for copying as much as it can, significantly reducing its costs and therefore increasing its margin and profit. (Table A2.3).

- N=3

Adding a new firm leads to an asymmetric situation with one monopoly and one duopoly. The market leader is the one with the highest starting quality, which engages in the most development effort as well, while its direct competitor cannot keep up with it and exerts a very small amount of development effort, leading to negligible profits with razor-thin margins. The monopolist on the second market performs significant amounts of development, close to those of the leader, and

reaches a high quality, awarding it significant profits with good margins, although still notably below the leader's. (Table A2.5).

Allowing for technology transfers leads to a very similar situation to Table A1.10 "One Market, 4 firms with reverse-engineering", with a leader harmed by the interaction; a runner-up (firm 3 here) which complements its reverse-engineering with a substantial amount of development effort, obtaining decent profits and margins; and a firm 2 which does not engage in original development, obtaining moderate profits with very high margins thanks to low copying costs. (Table A2.6).

- N=4

With 4 firms, the scenario recovers its symmetry. In the first market, the leader reaches the highest quality and profit as before, with a moderately high margin, while firm 2 forfeits its participation in development, obtaining negligible profits with only the starting quality. The runner-up, firm 3, takes advantage of its own market to develop a decent quality product, although not as high quality as the leader, and obtains a moderately high profit with a decent margin. Firm 4 does engage in a small amount of development, but it is not enough to compete with the other two giants and only manages to obtain very low profits with razor-thin margin. (Table A2.8).

Similarly, with reverse engineering, the market leader and the runner up (firms 1 and 3) are harmed and both reduce their final quality. Now, the newcomers (firms 2 and 4) do not engage in development effort and still focus on reverse-engineering, awarding them moderate profits, still below those of firm 3, but with very high margins, indicating the low costs of copying. (Table A2.9).

- N=6

With 6 firms, each market is composed of 3 firms. Here, the top 2 contenders in each market (firm 1 and 2 in the first market, and firms 4 and 5 in the second) engage in significant amounts of development, with profits largely dictated by their starting quality and moderate margins overall. Firm 5, with a modest starting quality, receives a small amount of profits with low margins, but still actively participates in the market. The remaining firms, 3 and 6, do not exert any development effort, obtaining negligible profits. (Table A2.11).

Again, here, the inclusion of reverse engineering allows most firms to obtain moderate profits with high margins by not exerting any development effort and just copying. Only firms 1, 2 and 4 venture into original development, with only the leader reaching a high final quality, while the qualities and profits of the other firms are largely comparable to each other. The profit and final quality of the leader are harmed by their competitor's copying, while its margin increases very slightly, likely attributable to lessened costs due to reduced development. All firms target the leader for its reverse-engineering efforts. (Table A2.12).

4.3 Results: Closed Silk Road

See appendix A3.

- N=3

With 3 firms, each having their own market, all firms engage in development efforts, achieving decent qualities. However, the leader's final quality, profit and margin overshadow the rest. Firm 3, due to being unreachable by the leader, exerts more development effort than firm 2, still without reaching the same level of quality, but finishing very close, and still obtains slightly higher profits than its rival, although with a lower margin. (Table A3.2).

In this case, allowing for reverse engineering does not modify the final qualities of the firms, nor the profits of the leader. It does, however, allow firms 2 and 3 to cut development costs and obtain higher profits and margins, leading to higher joint profits than before. (Table A3.3).

- N=5

In this run we have 5 firms divided into 3 markets: 2 duopolies connected only through a monopoly. We observe a strong market leader advantage in quality,

profit and margin, which progressively reduces as we advance through the firms. Notably, the 4th firm, belonging to the farthest market, is the second in development effort as well as profit. The last firm, despite a decent investment in development, obtains a low profit with a thin margin, due to its poor starting point. (Table A3.5).

As before, reverse engineering here does not damage the final quality nor the profits of the leader, but instead increases those of the rest of the firms, especially the last one. The joint profit is therefore increased. Here, since firms 4 and 5 do not have access to the first market, they copy instead from firm 3, until the maximum is reached. Notably, all firms do invest in original research nonetheless. (Table A3.6).

- N=8

With 4 successive duopolies, quality, profits and margins fall progressively as we advance through the network, with some exceptions. Firm 5, which is not connected to the leader but has a moderate head start, surpasses firm 4 in quality and profits, but not in margin. Similarly, firm 7 surpasses firm 6. This can be explained by the reduced level of quality in their local markets, which, coupled with their quality head start, allows them to reach higher qualities than their competitors, which are, however, not outstanding in the overall network. Interestingly, the average quality in the 4th farthest market is higher than in the 3rd market, which could possibly be explained by the reduced level of competition in the end nodes. Their margins, however, are still quite low. (Table A3.8).

Introducing reverse engineering only slightly modifies the picture. Although some harm is done to the market leader, the difference is very small. Improvements in the farthest firms are also subtle, with the greatest changes happening in firms 4 and 6. These are the weakest firms in their local markets, meaning they have more to catch-up from their more developed neighbors, especially for firm 4, which has access to the market leader: by shifting their focus from development to copying they are able to substantially increase their margins. Notably, firm 4 is the only one who manages to increase its final quality in a significant amount, with the others either decreasing it or staying roughly the same. As expected, all firms target the competitor with the highest starting quality to which they have access to, and always the maximum amount. (Table A3.9).

4.4 Results: Open Silk Road

See appendix A4.

- N=3

In this run we see a prominent market leader which takes most of the profit, but all firms achieve decent qualities and compete in the market. Notably, firm 3 exerts more development effort than firm 2, despite obtaining a lower final quality, given their poor starting point. Their margin is therefore quite low. (Table A4.2).

With reverse engineering, the final qualities of every firm, as well as the profit and margin of the leader, remain unchanged, while the profits and margins of firms 2 and 3 increase, especially for the latter. The change has not affected qualities but has indeed flattened profits. All firms copy from the leader until the maximum amount, despite the difference in costs and a higher joint profit is achieved. (Table A4.3).

- N=5

The addition of 2 more firms in the markets at the extremes weakens the market leader, but does not fare well for the newcomers either. Firm 2 now obtains a lower profit than firms 3 and 4 despite having higher margins, while firm 5 falls significantly short in quality and obtains a very small profit with only a thin margin. (Table A4.5).

Allowing reverse engineering greatly moderates these effects, although it does not negate them: the profits are flattened once more with minimal changes in final quality across the board. Margins are overall increased, especially for firm 5, which does not get involved in original development. Again, all firms copy the maximum amount from the leader regardless of distance. (Table A4.6).

- N=8

This is the most complex instance we analyzed, with a full network of 8 firms across 4 markets. In this run, we can observe how the firms with the highest head start in each market dominate over the less developed ones. A higher development effort combined with the given starting quality advantage allows them to obtain significantly greater profits than their direct competitors. Across the board, we can see that the leader maintains a significant advantage, while firms 4, 6 and 8 struggle to obtain significant profits, with markedly low margins. Notably, both

profits and margins fall overall as we descend through the table, with the exception of firms 5 and 7, which benefit from being the leaders of their local markets, as well as from limited competition due to being far away from the global leader. This effect is greater for firm 7 than for 5. (Table A4.8).

The inclusion of reverse engineering here does harm the quality of firms 1, 2 and 3 slightly, although the rest of the firms benefit from it. Only the leader, and also notably firm 7, reduces its profits, while the rest increases them. The firms which still engage in original development are 1, 2, 3 and only slightly 7. The rest simply copy from the leader until the maximum amount. Firms 4, 5 and 6 boast remarkably high margins, probably due to their lack of original research as well relative proximity to the leader, which allows them to copy for cheaper. (Table A4.9).

4.5 Results: Global Market

See appendix A5.

- N=3

Despite the imperfect links, the full network imposes strong competition between firms, heavily punishing the weakest contender with insignificant profit and a very thin margin. (Table A5.2).

With reverse-engineering, the last firm changes strategy and focuses on copying, getting it back into the market. Its profits are the lowest but still decent, and it has a very high margin. The top 2 firms are harmed by the change, in both quality and profit. Margins increase for firm 2 slightly but profits are reduced nonetheless. (Table A5.3).

- N=4

The newcomer, firm 4, is now driven out of the market without exerting any development effort, without any significant profit. Firm 3, however, does improve its position in every aspect. Although its profit and margin are not remarkable, they are now significant. (Table A5.5).

With the inclusion of reverse engineering, firms 3 and 4 forfeit their own original development and focus on copying, obtaining very similar profits, just under those of firm 2, and very wide margins. Firms 1 and 2 are harmed in terms of quality and

profit, although they maintain their dominant position in the market, especially firm 1 as the leader. (Table A5.6).

- $N=6$

We have a full network with 6 firms divided between 3 markets. Every second firm in each market fails to secure a spot for itself and is unable to exert development effort, obtaining no significant profit. The remaining 3 firms show a clear downward trend, where the lower their starting quality is, the lower their development efforts, profits and margin, yielding a notable advantage to the global leader. (Table A5.8).

With reverse engineering, only firms 1 and 3 engage in original development, where only firm 1 does so in a notable amount. All firms copy from the global leader until the maximum amount, obtaining the same final quality and almost identical profits and margins. The leader's final quality and profit are notably diminished, while joint profits are slightly increased. (Table A5.9).

5. Discussion of Results

5.1 Findings and Insights

1. Starting Quality

1. Firms with higher starting quality typically obtain a higher final quality, profit and often margin than their competitors.
 - This is well aligned with the literature. (Aghion et al, 2005; Gilbert, 2006)
2. Firms with low starting qualities can fail to develop in highly competitive markets.
 - This is also in agreement with the literature. (Aghion et al., 2005).

2. Competition

1. The profits and margins of the incumbent market leaders are typically reduced by the addition of new competitors.
 - This is a well-recognized effect of competition. (Dasgupta & Stiglitz, 1979).
2. Maximum final qualities are reduced by increasing competition.
 - While some sources support this finding (Boone, 2000), the majority of papers indicate that leading firms tend to increase their R&D efforts to escape competition. (Aghion et al., 2005; Dasgupta & Stiglitz, 1979; Gilbert, 2006).
3. Leaders in specific markets can perform well despite not being the global leader.
 - In agreement with the literature. (Gilbert, 2006).

3. Development effort

1. Market leaders, especially if they are monopolists, tend to exert the greatest development effort across all firms.
 - This is controversial. Chen & Schartz (2013) and Dasgupta & Stiglitz (1979) are in agreement with the statement, while Arrow (1962) and Boone (2000) are not.
2. When reverse engineering is allowed, non-leading firms tend to exert much less development effort.
 - Generally in agreement with Aghion et al. (2005) and Dasgupta & Stiglitz (1979), but not the exact focus of most models.

4. Reverse engineering

1. Leaders are unlikely to be harmed by non-competitors copying their technology.
 - While in agreement with basic theory, the model neglects the fact that technologies can be licensed to non-competing firms. Therefore, unlicensed use by non-competitors can indeed cause loss of income for IP holders. (Levin et al., 1987).
2. Firms always target the competitor with the highest starting quality they can reach, regardless of link strength.
 - This is a particularity of the model that does not have an accurate parallel in a real setting. It is likely related to how reverse-engineering costs are modelled.
3. Firms always reverse-engineer as much as they are allowed.
 - The literature states that reverse-engineering can be difficult and costly. (Levin et al., 1987; Moser, 2016). While the cost is included as a factor in the model, these results highlight that the cost of reverse engineering is likely not well represented.

5. Profits

1. Reverse engineering can increase joint profits, especially when firms with low starting qualities have at least some degree of market power due to network topology.
 - Some sources indicate that reverse engineering leads to increased competition, likely reducing profits. (Boldrin & Levine, 2013; Levin et al., 1987; Segerstrom, 1991). However, duplication of costs of R&D is a recognized inefficiency (Dasgupta & Stiglitz, 1979; Gilbert, 2006; Levin et al., 1987), which can be mitigated by reverse engineering or technology sharing.
2. The inclusion of reverse engineering tends to flatten the profit distribution by harming leaders and helping stragglers.
 - This finding is shared by other author's models. (Glass & Wu, 2007; Helpman, 1992; Levin et al., 1987).

6. Margins

1. Firms engaged in original development tend to have moderate margins.
 - Margins from R&D in the literature typically depend on the competition at the level of R&D facilities (laboratories). With free entry, the present value of profits tends towards zero (Dasgupta & Stiglitz, 1979; Segerstrom, 1991) or normal economic profits (Helpman, 1992).
2. Firms relying heavily, or entirely, on reverse engineering tend to achieve very wide margins, but limited profits.
 - In a competitive setting, imitators would obtain zero economic profits (Glass & Wu, 2007; Helpman, 1992). With limited entry, however, the limited profits would be split among the firms present (Segerstrom, 1991; Shaked & Sutton, 1982).

5.2 Answering the research questions

1. Under what conditions do firms exert effort in original development?

According to the model's results, firms exert effort in original development whenever they have, either an important advantage beforehand, or a significant amount of market power that can allow them to take advantage of the quality of the product. Firms which are leading at the start tend to exert significant effort to continue leading by the end of the cycle.

Firms that have maxed out their copying efforts sometimes choose to continue improving their product through original research if they have a certain degree of market power.

2. How does patent law infringement affect the quality of products in competitive markets?

Patent law infringements typically reduce the quality of the market leaders by reducing their incentives to exert effort in original development. It does, however, increase the quality of minor firms by allowing them to catch up with the leaders at little cost.

3. Would the elimination of patent protections increase efficiency?

Under some circumstances, the model indicates that joint profits can increase when reverse engineering occurs. This is due to the elimination of costly research duplicities among competing firms. However, such free riding and illegitimate competition also harm the incentives of the market leader to continue innovating, reducing the maximum quality reached. Minor firms do, however, increase their quality at little cost, which is potentially beneficial to low-end consumers.

6. Limitations

Among the many simplifications that were incorporated into the model, it is worthy to highlight the following:

Product development was only incorporated as a way to improve quality. In reality, firms often design their products in a way that differentiates them from their competition, allowing them to obtain higher margins. This could be shown as firms being able to modify their competition links to other firms, which the model does not allow in its current state.

One way in which products can be differentiated from each other is by establishing separate levels of quality aimed at different budgets. Although this unidimensional separation can be present in some of the tested scenarios, a critical part of this mechanism is consumer demand, which was not modelled in detail. In computing revenues, it was just assumed that more quality is always better, with disregard for the consumer budget. Without this crucial piece, strategic differentiation of product qualities cannot be understood to have taken place.

Another important point to consider is the unrealistic behavior shown by firms by always exerting the maximum amount of reverse engineering allowed. This is caused by the linear nature of the cost of reverse engineering, compared to the exponential nature of the cost of development. Given this circumstance, the cost parameter E can only act as a switch, turning the firm's ability to reverse-engineer on or off, without exerting a true modulating behavior.

Although it is common in the real world that the same firms that perform the research efforts are the ones to finally develop and manufacture the final goods, it is not always the case. Inventions and findings are often patented and sold to other companies. Sometimes, multiple manufacturing companies can license a single patent, making use of it simultaneously. However, this is not modelled here, and therefore it constitutes an important oversight in the way firms appropriate the proceeds of their R&D achievements. The markets for research activities, patentable inventions and final goods are three separate ones, which, in this model, have been fused together into one. This clearly limits the complexity of the interactions that can occur between those three related but separate equilibria.

Regarding the nature of R&D, it is important to remark that it tends to be a continuous and uncertain process. Firms may stumble upon significant findings by mere chance, or exert extenuating efforts without reaching their target. Moreover,

product developments tend to follow cycles, where a key discovery leads to a burst of new inventions, which critically slow down as the cycle reaches maturity. Incidentally, real-world firms can choose freely when and if they release new products, based on strategic considerations.

In this model, however, a single launch cycle is simulated, which is the same for all participants, and a single, exponentially increasing cost function. Each firm is assigned an arbitrary starting point, and an equilibrium is found for a single round. These facts make the model inappropriate for a true dynamic and quantitative analysis.

7. Conclusion

This work aimed to incorporate a competition analysis within the framework of a network model with the objective of clarifying the effects of different market structures in firm's R&D efforts, and how product imitation changes the dynamics of companies, compared to exclusively original research efforts.

The model, despite the simplicity of its components, showed significant realism, obtaining results coincident with the findings of similar models in the literature. Worthy of note are: that leading firms tend to maintain their advantage, that new firms often struggle in highly competitive markets, that firm's profits are reduced by the addition of new competitors, and that loosening intellectual property rights leads to a flattening of the profit distribution.

Its findings also contribute evidence to weigh in into important debates in the economics field. In particular, it indicated that development efforts are reduced when competition increases, that monopolists tend to exert the greatest development effort, and that the loosening of intellectual property rights can increase the joint profits of the industry by reducing duplicative costs.

However, caution is advised when interpreting these results at face value. An important limitation of the model was detected regarding the cost function of reverse engineering, which was found to not adequately represent real imitation dynamics. Moreover, another important caveat is that firms did not have the option to license their patents to other firms, and so their only way to profit from them was to participate in their direct commercialization. This oversight likely exaggerated the efficiency gains brought by reverse engineering. Such gains can also be replicated by paid patent licensing, which is a much fairer solution that respects intellectual property while keeping intact the incentives of innovators.

Despite the quantitative results obtained from the simulation, the model is only designed with qualitative aspects in mind. All equations, parameters and formulations were chosen for their simplicity, prioritizing tractability over realism and accuracy. Moreover, no real data was used to set up its parameters. Therefore, no quantitative or numerical conclusions should be drawn from it.

For the sake of expanding and refining the model, a few improvements are proposed. First, real historical data should be used to set up the model's equations and fine-tune its parameters, aiming to obtain quantitatively appropriate results. Second, the launch cycle should be simulated multiple times, using the end state

of one round as the starting point for the next, which would approach the model to a true system of dynamic competition. Finally, the markets for research capacity, patentable inventions, and manufactured goods should be simulated separately in order to obtain a more rich and realistic market behavior.

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Appendix A: Model output

A1: Single Market

Single Market: N=1

$$L_{ij} = 1$$

Firm	Target	Starting Quality (Q ₀)	Development effort (xd)	Final quality	Profit	Margin
1	1	1	99.0	149.0	9900	66.4%

Table A1.1: Nash Equilibrium of Scenario 1 with 1 firm

Single Market: N=2

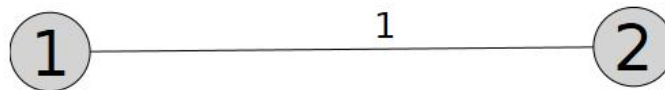


Figure A1.1: Network of Scenario 1 with 2 firms

$$L_{ij} =$$

1	1
1	1

Table A1.2: Link matrix of Single Market with 2 firms

Firm	Starting Quality (Q ₀)	Development effort (xd)	Final quality	Profit	Margin
1	50	95.2	145.2	7055	60.4%
2	1	34.3	35.3	68	9.8%

Table A1.3: Nash Equilibrium of Scenario 1 with 2 firms

Firm	Target	Starting Quality (Q ₀)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	92.2	142.2	6178	58.7%
2	1	1	49.0	0.0	50.0	1251	96.2%

Table A1.4: Nash Equilibrium of Scenario 1 with 2 firms and reverse-engineering

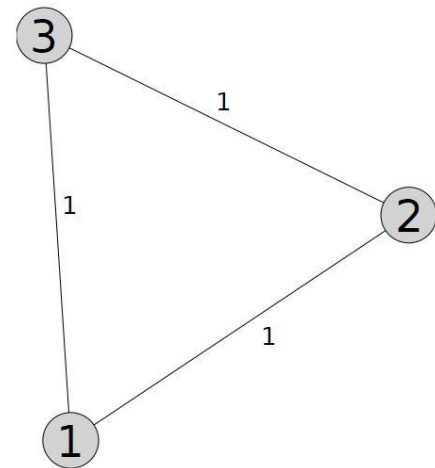
Single Market: N=3

Figure A1.2: Network of Scenario 1 with 3 firms

 $L_{ij} =$

1	1	1
1	1	1
1	1	1

Table A1.5: Link matrix of Scenario 1 with 3 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	83.1	133.1	4464	55.8%
2	25	62.3	87.3	1439	41.8%
3	1	0.0	1.0	0.45	100.0%

Table A1.6: Nash Equilibrium of Scenario 1 with 3 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	73.7	123.7	3353	54.6%
2	1	25	25.0	25.4	75.4	1274	55.9%
3	1	1	49.0	0.0	50.0	954	95.1%

Table A1.7: Nash Equilibrium of Scenario 1 with 3 firms and reverse-engineering

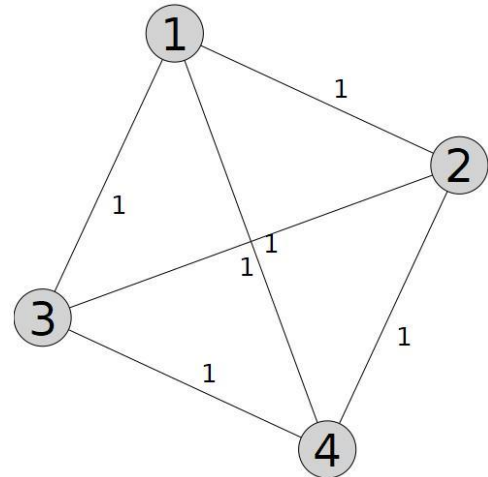
Single Market: N=4

Figure A1.3: Network of Scenario 1 with 4 firms

 $L_{ij} =$

1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1

Table A1.8: Link matrix of Scenario 1 with 4 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	75.9	125.9	3580	54.8%
2	25	53.0	78.0	1051	41.9%
3	10	27.6	37.6	174	29.9%
4	1	0.0	1.0	0.41	100.0%

Table A1.9: Nash Equilibrium of Scenario 1 with 4 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	64.0	114	2541	54.6%
2	1	25	25.0	15.3	65	987	64.7%
3	1	10	40.0	0.0	50	855	95.5%
4	1	1	49.0	0.0	50	846	94.5%

Table A1.10: Nash Equilibrium of Scenario 1 with 4 firms and reverse-engineering

A2: Two Markets

Two Markets: N=2

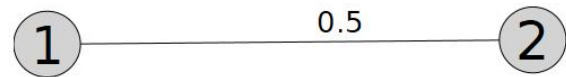


Figure A2.1: Network of Scenario 2 with 2 firms

$L_{ij} =$

1	0.5
0.5	1

Table A2.1: Link matrix of Scenario 2 with 2 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	94.6	144.6	6871	60.0%
2	1	75.3	76.3	1007	25.7%

Table A2.2: Nash Equilibrium of Scenario 2 with 2 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	94.6	144.6	6871	60.0%
2	1	1	49.0	26.3	76.3	2158	55.1%

Table A2.3: Nash Equilibrium of Scenario 2 with 2 firms and reverse-engineering

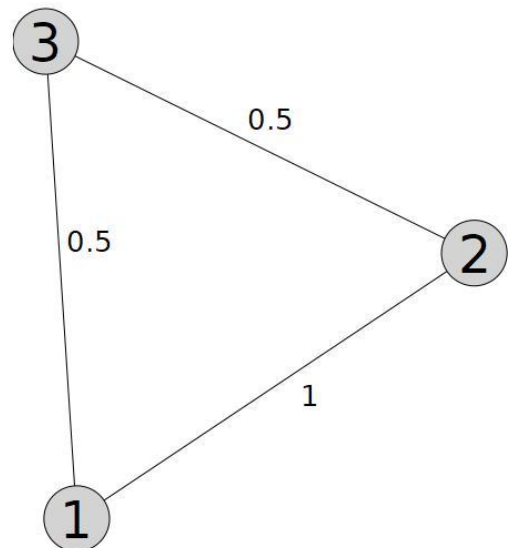
Two Markets: N=3

Figure A2.2: Network of Scenario 2 with 3 firms

 $L_{ij} =$

1	1	0.5
1	1	0.5
0.5	0.5	1

Table A2.4: Link matrix of Scenario 2 with 3 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	90.5	140.5	5778	58.0%
2	1	2.6	3.6	1	8.4%
3	25	83.0	108.0	2949	45.6%

Table A2.5: Nash Equilibrium of Scenario 2 with 3 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	79.9	129.9	4035	55.3%
2	1	1	49.0	0.0	50.0	1034	95.5%
3	1	25	25.0	52.1	102.1	2667	49.2%

Table A2.6: Nash Equilibrium of Scenario 2 with 3 firms and reverse-engineering

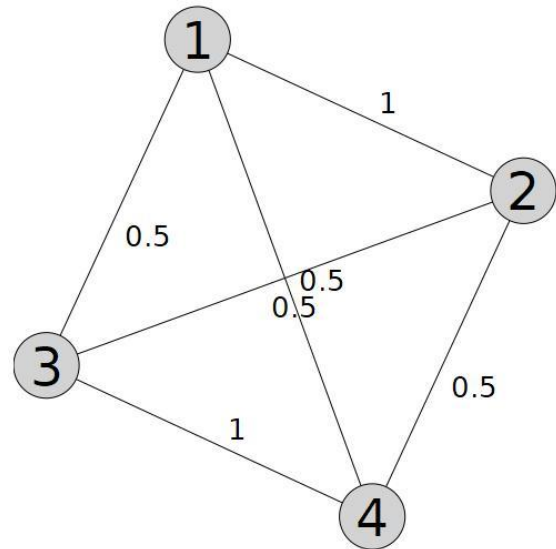
Two Markets: N=4

Figure A2.3: Network of Scenario 2 with 4 firms

 $L_{ij} =$

1	1	0.5	0.5
1	1	0.5	0.5
0.5	0.5	1	1
0.5	0.5	1	1

Table A2.7: Link matrix of Scenario 2 with 4 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	89.7	139.7	5602	57.7%
2	1	0.0	1.0	0.50	100.0%
3	25	77.7	102.7	2433	44.0%
4	1	16.7	17.7	8	4.9%

Table A2.8: Nash Equilibrium of Scenario 2 with 4 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	75.5	125.5	3540	54.8%
2	1	1	49.0	0.0	50.0	977	95.2%
3	1	25	25.0	36.1	86.1	1671	50.4%
4	1	1	49.0	0.0	50.0	1018	91.2%

Table A2.9: Nash Equilibrium of Scenario 2 with 4 firms and reverse-engineering

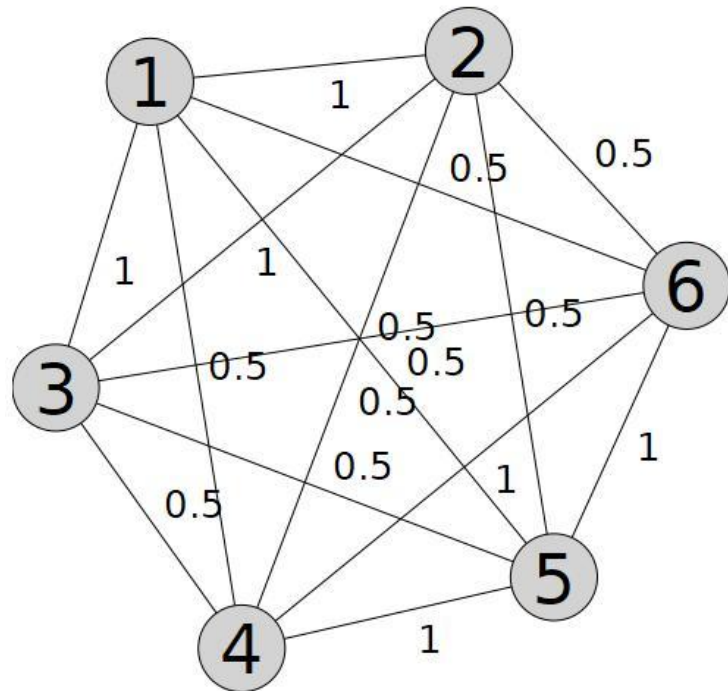
Two Markets: N=6

Figure A2.4: Network of Scenario 2 with 6 firms

 $L_{ij} =$

1	1	1	0.5	0.5	0.5
1	1	1	0.5	0.5	0.5
1	1	1	0.5	0.5	0.5
0.5	0.5	0.5	1	1	1
0.5	0.5	0.5	1	1	1
0.5	0.5	0.5	1	1	1

Table A2.10: Link matrix of Scenario 2 with 6 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	70.5	120.5	3063	54.5%
2	25	46.9	71.9	853	42.7%
3	1	0.0	1.0	0.39	100.0%
4	25	59.8	84.8	1324	41.7%
5	10	34.2	44.2	242	28.2%
6	1	0.0	1.0	0.44	100.0%

Table A2.11: Nash Equilibrium of Scenario 2 with 6 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	57.8	107.8	2134	55.2%
2	1	25	25.0	9.8	59.8	860	72.4%
3	1	1	49.0	0.0	50.0	781	94.1%
4	1	25	25.0	16.5	66.5	991	61.8%
5	1	10	40.0	0.0	50.0	828	91.2%
6	1	1	49.0	0.0	50.0	810	89.2%

Table A2.12: Nash Equilibrium of Scenario 2 with 6 firms and reverse-engineering

A3: Closed Silk Road

Closed Silk Road: N=3

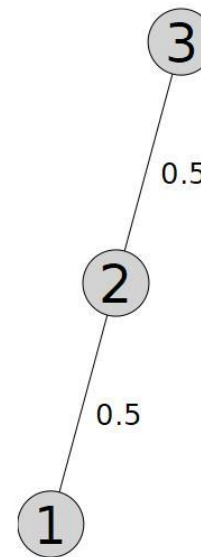


Figure A3.1: Network of Scenario 3 with 3 firms

$L_{ij} =$

1	0.5	0
0.5	1	0.5
0	0.5	1

Table A3.1: Link matrix of Scenario 3 with 3 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	92.9	142.9	6360	59.0%
2	25	68.5	93.5	1769	42.3%
3	1	87.0	88.0	1876	32.7%

Table A3.2: Nash Equilibrium of Scenario 3 with 3 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	92.9	142.9	6360	59.0%
2	1	25	25.0	43.5	93.5	2057	49.2%
3	2	1	24.0	63.0	88.0	2140	37.3%

Table A3.3: Nash Equilibrium of Scenario 3 with 3 firms and reverse-engineering

Closed Silk Road: N=5

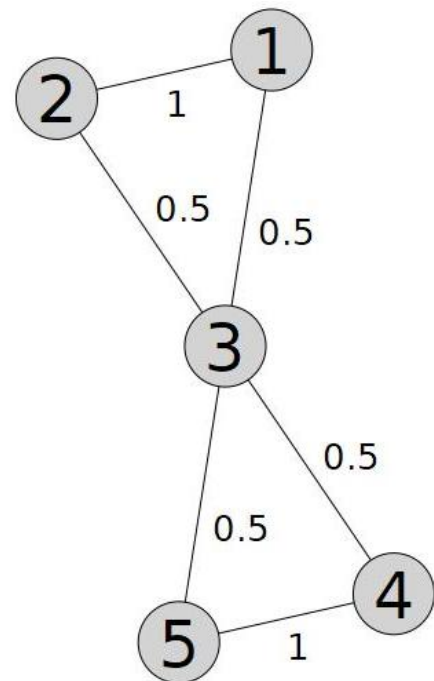


Figure A3.2: Network of Scenario 3 with 5 firms

 $L_{ij} =$

1	1	0.5	0	0
1	1	0.5	0	0
0.5	0.5	1	0.5	0.5
0	0	0.5	1	1
0	0	0.5	1	1

Table A3.4: Link matrix of Scenario 3 with 5 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	75.9	125.9	3583	54.8%
2	25	53.0	78.0	1053	41.9%
3	25	52.0	77.0	1016	42.0%
4	10	72.0	82.0	1278	32.4%
5	1	49.1	50.1	217	14.7%

Table A3.5: Nash Equilibrium of Scenario 3 with 5 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	75.9	125.9	3583	54.8%
2	1	25	25.0	28.0	78.0	1365	54.4%
3	1	25	25.0	27.0	77.0	1303	53.9%
4	3	10	15.0	57.0	82.0	1375	34.9%
5	3	1	24.0	25.1	50.1	481	32.7%

Table A3.6: Nash Equilibrium of Scenario 3 with 5 firms and reverse-engineering

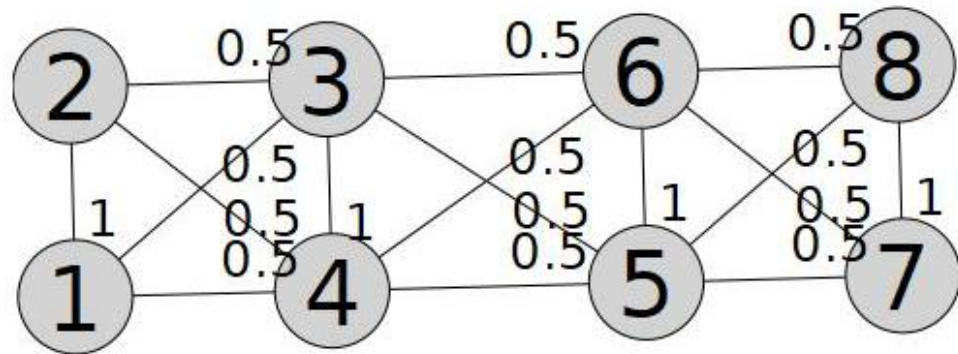
Closed Silk Road: N=8

Figure A3.3: Network of Scenario 3 with 8 firms

 $L_{ij} =$

1	1	0.5	0.5	0	0	0	0
1	1	0.5	0.5	0	0	0	0
0.5	0.5	1	1	0.5	0.5	0	0
0.5	0.5	1	1	0.5	0.5	0	0
0	0	0.5	0.5	1	1	0.5	0.5
0	0	0.5	0.5	1	1	0.5	0.5
0	0	0	0	0.5	0.5	1	1
0	0	0	0	0.5	0.5	1	1

Table A3.7: Link matrix of Scenario 3 with 8 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	72.8	122.8	3274	54.6%
2	25	49.4	74.4	932	42.3%
3	25	49.1	74.1	920	42.3%
4	10	24.4	34.4	147	31.3%
5	10	45.6	55.6	412	27.5%
6	5	30.2	35.2	114	19.0%
7	5	64.6	69.6	727	25.3%
8	1	52.3	53.3	268	15.9%

Table A3.8: Nash Equilibrium of Scenario 3 with 8 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	71.6	121.6	3162	54.5%
2	1	25	25.0	23.1	73.1	1202	57.4%
3	1	25	25.0	20.8	70.8	1109	57.9%
4	1	10	40.0	0.0	50.0	876	91.6%
5	3	10	15.0	28.7	53.7	476	34.5%
6	3	5	20.0	8.2	33.2	280	53.4%
7	5	5	5.0	60.2	70.2	754	25.7%
8	5	1	9.0	44.1	54.1	314	18.0%

Table A3.9: Nash Equilibrium of Scenario 3 with 8 firms and reverse-engineering

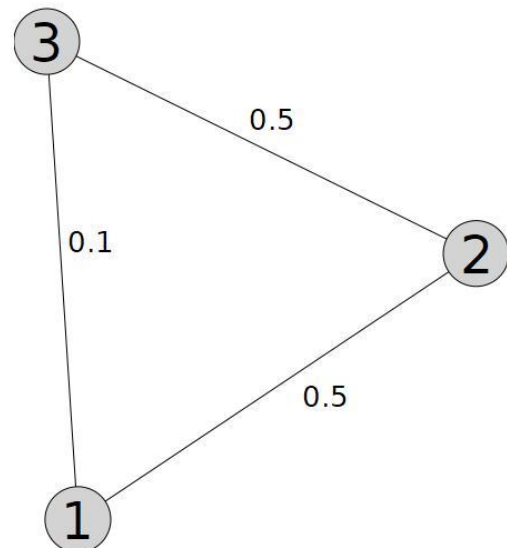
Open Silk Road: N=3

Figure A4.1: Network of Scenario 4 with 3 firms

 $L_{ij} =$

1	0.5	0.1
0.5	1	0.5
0.1	0.5	1

Table A4.1: Link matrix of Scenario 4 with 3 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	91.0	141.0	5882	58.2%
2	25	69.9	94.9	1856	42.5%
3	1	80.5	81.5	1323	28.5%

Table A4.2: Nash Equilibrium of Scenario 4 with 3 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	91.0	141.0	5882	58.2%
2	1	25	25.0	44.9	94.9	2143	49.1%
3	1	1	49.0	31.5	81.5	2083	44.9%

Table A4.3: Nash Equilibrium of Scenario 4 with 3 firms and reverse-engineering

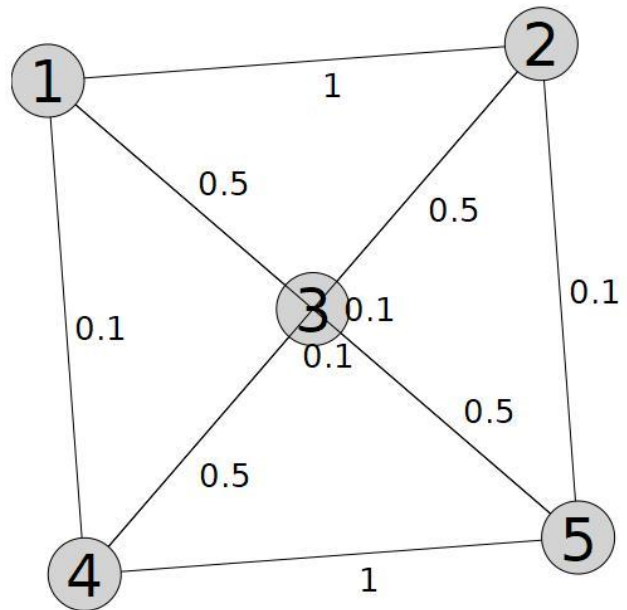
Open Silk Road: N=5

Figure A4.2: Network of Scenario 4 with 5 firms

 $L_{ij} =$

1	1	0.5	0.1	0.1
1	1	0.5	0.1	0.1
0.5	0.5	1	0.5	0.5
0.1	0.1	0.5	1	1
0.1	0.1	0.5	1	1

Table A4.4: Link matrix of Scenario 4 with 5 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	73.3	123.3	3321	54.6%
2	25	50.0	75.0	950	42.2%
3	25	54.9	79.9	1123	41.8%
4	10	67.1	77.1	1035	30.9%
5	1	39.4	40.4	105	11.4%

Table A4.5: Nash Equilibrium of Scenario 4 with 5 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	73.3	123.3	3319	54.6%
2	1	25	25.0	25.0	75.0	1261	56.1%
3	1	25	25.0	29.1	79.1	1380	52.9%
4	1	10	40.0	23.1	73.1	1315	44.9%
5	1	1	49.0	0.0	50.0	880	64.2%

Table A4.6: Nash Equilibrium of Scenario 4 with 5 firms and reverse-engineering

Open Silk Road: N=8 $L_{ij} =$

1	1	0.5	0.5	0.2	0.2	0.05	0.05
1	1	0.5	0.5	0.2	0.2	0.05	0.05
0.5	0.5	1	1	0.5	0.5	0.2	0.2
0.5	0.5	1	1	0.5	0.5	0.2	0.2
0.2	0.2	0.5	0.5	1	1	0.5	0.5
0.2	0.2	0.5	0.5	1	1	0.5	0.5
0.05	0.05	0.2	0.2	0.5	0.5	1	1
0.05	0.05	0.2	0.2	0.5	0.5	1	1

Table A4.7: Link matrix of Scenario 4 with 8 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	69.2	119.2	2950	54.5%
2	25	45.5	70.5	813	42.9%
3	25	46.9	71.9	852	42.7%
4	10	22.7	32.7	133	32.2%
5	10	38.0	48.0	291	27.7%
6	5	22.6	27.6	69	20.0%
7	5	58.1	63.1	529	23.2%
8	1	42.7	43.7	137	12.5%

Table A4.8: Nash Equilibrium of Scenario 4 with 8 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	67.2	117.2	2790	54.5%
2	1	25	25.0	18.5	68.5	1070	61.2%
3	1	25	25.0	15.4	65.4	965	62.9%
4	1	10	40.0	0.0	50.0	816	91.1%
5	1	10	40.0	0.0	50.0	815	80.3%
6	1	5	45.0	0.0	50.0	790	77.8%
7	1	5	45.0	3.0	53.0	473	31.2%
8	1	1	49.0	0.0	50.0	369	27.3%

Table A4.9: Nash Equilibrium of Scenario 4 with 8 firms and reverse-engineering

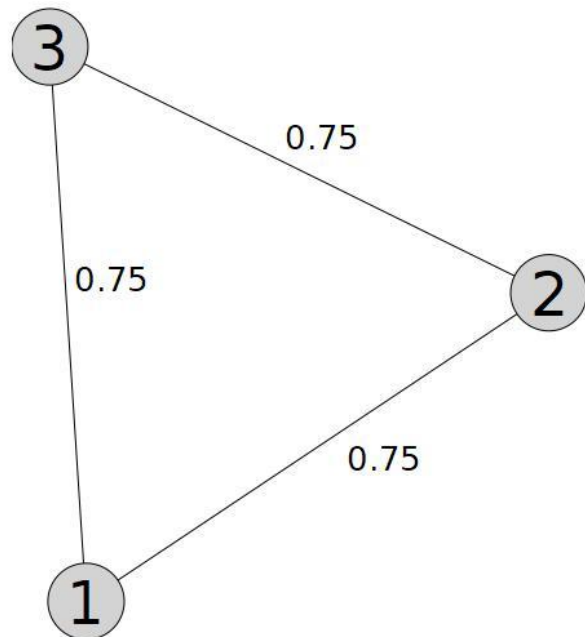
Global Market: N=3

Figure A5.1: Network of Scenario 5 with 3 firms

 $L_{ij} =$

1	0.75	0.75
0.75	1	0.75
0.75	0.75	1

Table A5.1: Link matrix of Scenario 5 with 3 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	84.0	134.0	4602	56.0%
2	25	68.6	93.6	1781	42.3%
3	1	18.2	19.2	10	5.3%

Table A5.2: Nash Equilibrium of Scenario 5 with 3 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	79.4	129.4	3977	55.2%
2	1	25	25.0	37.2	87.2	1737	50.7%
3	1	1	49.0	0.0	50.0	1112	94.4%

Table A5.3: Nash Equilibrium of Scenario 5 with 3 firms and reverse-engineering

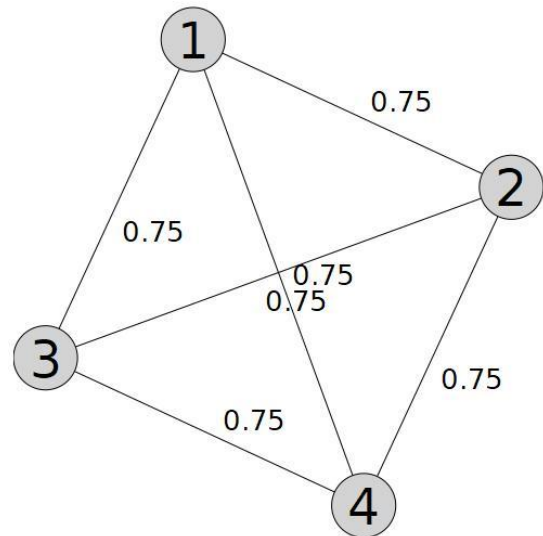
Global Market: N=4

Figure A5.2: Network of Scenario 5 with 4 firms

 $L_{ij} =$

1	0.75	0.75	0.75
0.75	1	0.75	0.75
0.75	0.75	1	0.75
0.75	0.75	0.75	1

Table A5.4: Link matrix of Scenario 5 with 4 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	79.1	129.1	3938	55.1%
2	25	61.7	86.7	1412	41.8%
3	10	41.1	51.1	336	27.5%
4	1	0.0	1.0	0.50	100.0%

Table A5.5: Nash Equilibrium of Scenario 5 with 4 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	71.7	121.7	3170	54.5%
2	1	25	25.0	27.5	77.5	1337	54.3%
3	1	10	40.0	0.0	50.0	1002	94.9%
4	1	1	49.0	0.0	50.0	990	93.8%

Table A5.6: Nash Equilibrium of Scenario 5 with 4 firms and reverse-engineering

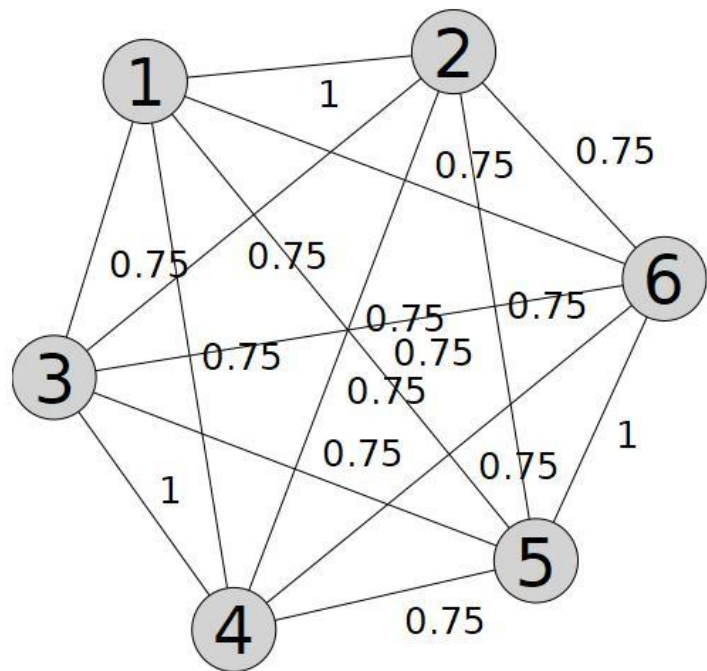
Global Market: N=6

Figure A5.3: Network of Scenario 5 with 6 firms

 $L_{ij} =$

1	1	0.75	0.75	0.75	0.75
1	1	0.75	0.75	0.75	0.75
0.75	0.75	1	1	0.75	0.75
0.75	0.75	1	1	0.75	0.75
0.75	0.75	0.75	0.75	1	1
0.75	0.75	0.75	0.75	1	1

Table A5.7: Link matrix of Scenario 5 with 6 firms

Firm	Starting Quality (Q0)	Development effort (xd)	Final quality	Profit	Margin
1	50	78.8	128.8	3904	55.1%
2	1	0.0	1.0	0.43	100.0%
3	25	61.4	86.4	1394	41.8%
4	1	0.0	1.0	0.45	100.0%
5	10	40.6	50.6	329	27.5%
6	1	0.0	1.0	0.47	100.0%

Table A5.8: Nash Equilibrium of Scenario 5 with 6 firms

Firm	Target	Starting Quality (Q0)	Reverse-engineering effort (xe)	Development effort (xd)	Final quality	Profit	Margin
1	1	50	0.0	54.9	104.9	1962	55.7%
2	1	1	49.0	0.0	50.0	751	93.9%
3	1	25	25.0	9.8	59.8	851	71.7%
4	1	1	49.0	0.0	50.0	765	92.1%
5	1	10	40.0	0.0	50.0	784	93.6%
6	1	1	49.0	0.0	50.0	772	92.2%

Table A5.9: Nash Equilibrium of Scenario 5 with 6 firms and reverse-engineering